JOURNAL OF INQUIRY-BASED LEARNING IN MATHEMATICS
No. **33**, (Oct. 2013)

Foundations of Analysis[1]

W. Ted Mahavier << W.S. Mahavier

Lamar University
Emory University

[1]If each of X and Y is an author, then the relation X << Y indicates that these notes were originally authored by Y and subsequently prepared for submission to JIBLM by X without significant modification and with appropriate permissions. While Y is credited with the genesis of the notes, X takes full responsibility for the current version.

Contents

To the Student

Analysis is an area of mathematics, just as Algebra, Geometry, and Topology are areas of mathematics, and is usually defined heuristically or not at all. In your calculus sequence you learned about the topics of limits, continuity, differentiability, and integration at a computational level. In this course, we will follow the same order that you followed in calculus, but we will spend more time on the mathematical structures than on the application of the concepts. We will define each concept rigorously and present material that you will recognize from calculus such as the Extreme Value Theorem and the Fundamental Theorem of Calculus.

These notes are intended to be self-contained, only assuming a basic understanding of the real numbers. If you work through these problems independently, then you will have mastered the topic that most universities would cover in an undergraduate course titled “Real Analysis” or “Advanced Calculus.” Universities typically offer such a course after a “transitional” course intended to move students away from working problems and toward a theoretical viewpoint of the subject. This course serves the purpose of both courses by using the ideas of Analysis to help you make this transition.

All work presented or submitted is to be your own. You are *not* to discuss any problem with any one other than me, nor are you to look to any other reference such as a book or the internet for further guidance.

Grading for the course will be no less than the average of three grades: your presentation grade, your submission grade, and the average of your midterm and final exam grades. Anyone who is regularly presenting material at the board will certainly have adequate work for good submission grades and thus will likely do well on the midterm and final. The midterm and final grades are opportunities for those who do not regularly make it to the board. However, it is my experience that those who do not work toward successful presentations rarely do well on the midterm and final. Thus, I emphasize that your *goal* for the course should be well prepared, well presented problems at the board. You will know your grades on submissions and tests. My policy for your presentation grade follows.

D = You made it to class every day, were attentive and alert.

C = You met the requirements for D plus made a few successful presentations.

B = You met the requirements for C plus made numerous successful presentations.

A = You met the requirements for B plus presented some truly impressive problems.

Turn-ins. You must turn in exactly one "new" problem each week. A "new" problem means one that you have *not* turned in before. This problem should be neatly written and double spaced. You should label this problem with TURN IN at the top of the page along with your name, the problem number and the problem statement.

Grading for TURN IN assignments will be based on this scale.

A = You turned in a correct proof.

B = You know how to prove the theorem but some of what you have written is not correct.

C = You have a mistake in your work or I do not understand what you have written, but I believe that you have a good idea.

D = There is at least one major flaw in your argument.

The purpose of the TURN IN assignments is to *train* you to prove theorems. It is not expected that you started the class with this skill. Hence, some low grades are to be expected. Do not be upset. Just come see me.

Resubmissions. If you receive a grade of less than B, you may resubmit a TURN IN problem on the following week and I will average the two grades. You still must turn in a new problem as well and you only get one chance to resubmit. Please write RESUBMIT at the top. Feel free to come see me anytime if you do not understand my comments. It is expected that a certain amount of time in my office will be required. I prefer to give guidance in my office rather than in class because this allows me to tailor the guidance to you. Of course, you should still feel free to ask questions in class, even though I may ask you to stop by the office to discuss a particular question.

Boardwork. If you have a solution to a problem that is about to be presented, you may turn that problem in for boardwork credit, although it receives less credit than a presented problem. If you have solved a problem that is about to be presented at the board *or* you have made significant progress on a problem that is about to be presented then you may opt to leave the room for the presentation. In this case, you may turn in a write up of this problem for credit as BOARD WORK. You must write BOARD WORK at the top of the page. There is no limit on the number of BOARD WORK problems you may submit.

Last Comments. Be sure that everything you turn in has your name, problem number, and problem statement on it. Be sure to double space and write either TURN IN, RESUBMIT, or BOARD WORK at the top.

I reserve the right to vary from these policies and probably will.

Problem Sequence

I prefer geometric language and I will use the words point and number interchangeably, interpreting the word point to mean a point on the number line. For example if a and b are two numbers and I say that a is to the left of b then I mean that $a < b$. You should feel free to say $a < b$ or a is to the left of b, as you prefer. Similarly you may say $a > b$ or a is to the right of b.

Definition 1. *If a and b are two points and a is to the left of b, the statement that the point p is **between** the points a and b means that p is to the right of a and to the left of b.*

The following is a list of a few basic properties of the number line. You should explain what properties you are using when working a problem. You do not have to restrict yourself to this list but may use any property of the numbers that you know. We will use these properties on many of our first problems.

Axioms of the number line

Axiom 1. *If each of a and b is a number then exactly one of the following is true: a is b, a is to the left of b, or a is to the right of b.*

Axiom 2. *If each of a, b, and c is a number, and a is to the left of b, and b is to the left of c, then a is to the left of c.*

Axiom 3. *If a and b are two points on the number line there is a point between them.*

Axiom 4. *If a is a point, there is a smallest integer to the right of a and a largest integer to the left of a.*

Axiom 5. *If n is an integer then $n-1$ and $n+1$ are integers and n is the only integer between $n-1$ and $n+1$.*

Definition 2. *A **point set** is a set of one or more points.*

In some courses it is convenient to introduce what is called the empty set, a set that contains no points (or anything else). In this class we will assume that all our point sets are not empty and thus contain at least one point. We will also assume that when we write "a and b are two points," that we mean that a and b are not equal.

Definition 3. *The statement that the point set S is a **segment** means that there are two points a and b, called the endpoints of S, such that S is the set of all points between a and b.*

Notation 1. *If a and b are two points and $a < b$ then (a,b) denotes the segment consisting of all points between a and b.*

A segment is a special kind of point set but in the definition of a limit point that follows you should not assume that M is a segment. You should assume only that M is a collection of one or more points.

Definition 4. *If M is a point set and p is a point, the statement that p is a **limit point** of the point set M means that* every *segment that contains p contains a point of M different from p.*

Note that the definition does not say that the point p is or is not in M.

Problem 1. *Show that if M is the set which contains only the number 0, and p is a point, then p is not a limit point of M.*

Problem 2. *Show that if M is the point set that contains only the two points 0 and 1, then no point is a limit point of M.*

Problem 3. *Show that 1 is a limit point of the segment $(0,1)$.*

Problem 4. *Show that a and b are limit points of the segment (a,b).*

Problem 5. *If S is the segment (a,b), show that every point of S is a limit point of S.*

Definition 5. *The statement that the point set I is an **interval** means that there are two points a and b, called the endpoints of I, such that I is the set containing a, b, and all points between a and b.*

Notation 2. *If a and b are two points and $a < b$ then $[a,b]$ denotes the interval with endpoints a and b.*

Problem 6. *Show that if M is an interval and p is a point not in M, then p is not a limit point of M.*

Definition 6. *The statement that the point set H is a **subset** of the point set K means that if p is a point of H, then p is a point of K.*

Notation 3. *If H is a point set and K is a point set then $H \subseteq K$ means that H is a subset of K.*

For the next problem you are not to assume anything about H and K except what you are given, namely that each of H and K is a set of one or more points, and $H \subseteq K$, and that p is a limit point of H.

Problem 7. *Show that if H is a point set, and K is a point set, and $H \subseteq K$, and p is a limit point of H, then p is a limit point of K.*

Problem 8. *Show that if M is the set of all positive integers and p is a point, then p is not a limit point of M.*

Problem 9. *Show that if p is a limit point of the point set M and S is a segment containing p, then S contains 2 points of M.*

Problem 10. *Let H be a point set which has a limit point, and let K be the set of all limit points of H. Show that if p is a limit point of K, then p is also a limit point of H.*

Problem 11. *Show that if M is the set of all reciprocals of positive integers, then 0 is a limit point of M.*

Problem 12. *Show that if $p \neq 0$, then p is not a limit point of the set of all reciprocals of positive integers.*

Definition 7. *If each of H and K is a point set and there is a point that is in both of them, then the **intersection** of H and K is the set to which a point p belongs if and only if p is in both H and K.*

Notation 4. *If each of H and K is a point set and there is a point that is in both of them, then $H \cap K$ denotes the intersection of H and K.*

Problem 13. *If H and K are two point sets having a common point, and p is a limit point of $H \cap K$, then p is a limit point of H and p is a limit point of K.*

Problem 14. *Suppose that M is a point set and p is a limit point of M. Must it be true that every interval containing p contains a point of M different from p?*

Problem 15. *Suppose that M is a point set and every interval containing p contains a point of M different from p. Must it be true that p is a limit point of M?*

Definition 8. *If each of H and K is a point set the **union** of H and K is the set to which the point p belongs if and only if p is in H or p is in K.*

Notation 5. *If each of H and K is a point set, then $H \cup K$ denotes the union of H and K. Thus $H \cup K$ is the set of all points in H together with the points in K.*

Problem 16. *If the point p is in each of the two segments S_1 and S_2, then $S_1 \cap S_2$ is a segment containing p.*

Problem 17. *Show that if p is not a limit point of the point set H and p is not a limit point of the point set K, then p is not a limit point of $H \cup K$.*

Problem 18. *Show that if H and K are two point sets and p is a limit point of $H \cup K$, then p is a limit point of H or p is a limit point of K.*

Notation 6. *If each of a,b, and c is a number, we will use the notation $M = \{a,b,c\}$ to mean the set containing the points a, b, and c and no other point. Similarly we will denote infinite sets, when the pattern is clear, by $M = \{a_1,a_2,a_3,\ldots\}$. For example, the set of all positive integers is denoted by $M = \{1,2,3,\ldots\}$.*

Definition 9. *The statement that the point set M is* ***infinite*** *means that for every positive integer n, M contains (at least) n points.*

Problem 19. *Show that if M is a point set and there is no smallest number in M (or there is no largest number in M), then if n is a positive integer, M has n points so that M is infinite.*

Definition 10. *The statement that the point set M is finite means that it is not infinite. That is, there is a positive integer n such that M does not contain n points.*

Problem 20. *Show that if M is a point set and M is finite then there is a smallest (and a largest) number in M.*

Problem 21. *Show that if M is a point set, and p is a limit point of M, and S is a segment containing p, and n is a positive integer, then S contains n points so that $S \cap M$ is infinite.*

By a **function** we mean a set of ordered number pairs, no two of which have the same first term. Or, if you prefer, a function is a set of points in the number plane with no two on the same vertical line. If f is a function, then the **domain** of f is the set of all first terms of ordered pairs of f and the **range** of f is the set of all second terms of ordered pairs of f. By a **sequence** we mean a function whose domain is the set of positive integers and whose range is a point set.

Usually if f is a function and (x,y) is one of the ordered pairs in f, then we denote y by $f(x)$. When f is a sequence and (n,y) is one of the ordered pairs in f, then we usually denote y by the short hand f_n. Thus we might refer to a sequence by the name of the function as f for example. Or we might refer to a sequence as $f_1, f_2, f_3, \ldots$. By a term of a sequence f, or a point of the sequence, or a point in the sequence, we mean f_n for some positive integer n.

Definition 11. *The statement that the sequence $x_1,x_2,x_3,\ldots$* ***converges*** *to the number c or has c as a* ***limit*** *means that for every segment S containing c, there is a positive integer n such that each of $x_n,x_{n+1},x_{n+2},\ldots$ is in S.*

If n is a positive integer, another way of saying that each of $x_n,x_{n+1},x_{n+2},\ldots$ is in (a,b) is to say that for each integer m such that $m \geq n$, x_m is in (a,b).

Problem 22. *What does it mean to say that a sequence $x_1,x_2,x_3,\ldots$ does not converge to the number c?*

Problem 23. *Show that the sequence* $0,1,0,1,...$ *does not converge to* 0.

Problem 24. *Show that the sequence* $1,\frac{1}{2},\frac{1}{3},\frac{1}{4},\dots$ *converges to* 0.

Problem 25. *Show that if the sequence* $x_1,x_2,x_3,...$ *converges to the number* c *and* (a,b) *is a segment containing* c *then there are not infinitely many terms of the sequence* $x_1,x_2,x_3,...$ *that are not in* (a,b).

Problem 26. *Does the sequence* $\frac{1}{2},\frac{1}{1},\frac{1}{4},\frac{1}{3},\frac{1}{6},\frac{1}{5},\dots$ *converge to* 0*? Note that this is the sequence* x *where* $x_n=\frac{1}{n-1}$ *if* n *is an even positive integer and* $x_n=\frac{1}{n+1}$ *if* n *is an odd positive integer.*

Problem 27. *Assume that the sequence* $x_1,x_2,x_3,...$ *converges to the point* c *and* d *is a point different from* c*. Show that* $x_1,x_2,x_3,...$ *does not converge to* d.

Note that a sequence is not a point set, but a point of ordered pairs. As such, it does not have a limit point. However, the range of a sequence is a point set and thus might or might not have a limit point.

Problem 28. *Show that if the sequence* $x_1,x_2,x_3,...$ *converges to the point* c*, and, for each positive integer* n*,* $x_n \neq x_{n+1}$*, then* c *is a limit point of the range of the sequence.*

Problem 29. *Find a sequence* $x_1,x_2,x_3,\dots$ *such that the point* c *is a limit point of the range* $\{x_1,x_2,\dots\}$ *of the sequence but* $x_1,x_2,x_3,\dots$ *does not converge to* c.

Problem 30. *Show that if* d *is a number and* $x_1,x_2,x_3,...$ *is a sequence which converges to the point* c*, then the sequence* $d\cdot x_1, d\cdot x_2, d\cdot x_3,...$ *converges to* $d\cdot c$.

Problem 31. *Find examples of infinite sets, one having a first point and one not having a first point.*

Problem 32. *Assume that* $x_1,x_2,x_3,...$ *is a sequence that converges to the point* c *and* d *is a point different from* c*. Show that* d *is not a limit point of the range of the sequence.*

Problem 33. *Show that if* M *is a point set, there cannot be both a rightmost point of* M *and a smallest number which is larger than each number in* M.

Problem 34. *If* M *is a point set and each point of* M *is to the left of* p *and* p *is the smallest number such that each point of* M *is to the left of* p*, then* p *is a limit point of* M.

Definition 12. *The statement that the point set* M *is* ***bounded above*** *means that there is a number* c *such that each point of* M *is to the left of* c.

Definition 13. *The statement that the point set* M *is* ***bounded below*** *means that there is a number* c *such that each point of* M *is to the right of* c.

Notation 7. *If p is a point and M is a point set, then $p \in M$ means that p is a point in M. Similarly $p \notin M$ means that p is not in M.*

Axiom 6. ***Completeness Axiom*** *If M is a point set which is bounded above (below) , then there is a point p such that either*
1) $p \in M$ and p is the largest (smallest) number in M, or
2) $p \notin M$ and p is the smallest (largest) number such that each point of M is to the left (right) of p.

Definition 14. *The statement that the sequence $x_1, x_2, x_3, ...$ is an **increasing (decreasing)** sequence means that for each positive integer n, $x_n < x_{n+1}$ ($x_n > x_{n+1}$).*

Problem 35. *If the range of the increasing sequence $x_1, x_2, x_3, \ldots$ is bounded above, then $x_1, x_2, x_3, ...$ converges to some point.*

Problem 36. *Assume that $x_1, x_2, x_3, ...$ is a sequence and ε is a positive number and S is the segment $(c-\varepsilon, c+\varepsilon)$. Show that there is a positive integer n such that $\{x_n, x_{n+1}, x_{n+2} \ldots\} \subseteq S$. Show that $x_1, x_2, x_3, \ldots$ converges to c.*

Problem 37. *Show that if the sequence $p_1, p_2, p_3, ...$ converges to c and the sequence $q_1, q_2, q_3, ...$ converges to d, then the sequence $p_1 + q_1, p_2 + q_2, p_3 + q_3, \ldots$ converges to $c + d$.*

Definition 15. *The statement that the sequence $x_1, x_2, x_3, ...$ is **non decreasing (non increasing)** means that for each positive integer n, $x_n \leq x_{n+1}$ ($x_n \geq x_{n+1}$).*

Problem 38. *If the range of the non decreasing sequence $x_1, x_2, x_3, ...$ is bounded above, then $x_1, x_2, x_3, ...$ converges to some point.*

Definition 16. *The statement that the point set M is **bounded** means that M is bounded above and M is bounded below. Equivalently, there is a segment that contains M.*

Definition 17. *If M is a point set that is bounded above then the **least upper bound** of M is the number that is the largest number in M or the smallest number that is larger than each number in M. It is denoted by $lub(M)$.*

Definition 18. *If M is a point set that is bounded below then the **greatest lower bound** of M is the number that is the smallest number in M or the largest number that is smaller than each number in M. It is denoted by $glb(M)$.*

Problem 39. *If M is a bounded point set then the $lub(M)$ (and the $glb(M)$) is either a point of M or a limit point of M.*

Problem 40. *Give an example of a bounded point set such that lub(M) is both a point of M and a limit point of M.*

Problem 41. *If the sequence $x_1, x_2, x_3, \ldots$ converges to the point c, then $M = \{x_1, x_2, x_3 \ldots\}$ is bounded.*

Problem 42. *If M is an infinite and bounded point set, then M has a limit point.*

Note 1. *In the following, we will be talking about points in a plane. We will denote them by their coordinates as (x,y). You will have to determine from the context whether (x,y) denotes a segment from x to y, or the point with coordinates x and y. We will also have to be careful to distinguish between* points *and* points on the x-axis *which are really points of the form $(x,0)$ but which we often think of as numbers.*

Definition 19. *The statement that f is a **simple graph** (or a **function**) means that f is a set of points (x,y) in a plane such that no vertical line contains two of them.*

If f is a function then the set of first terms of the points of f is called the domain of f and the set of second terms of the points of f is called the range of f. We use the usual notation that if x is a number in the domain of f, then $f(x)$ is the number which is the second coordinate of the point of f whose first coordinate is x.

Definition 20. *The statement that the simple graph f is **continuous** at the point $(x, f(x))$ means that if S is a (any) segment containing $f(x)$ then there is a segment T containing x such that if $t \in T$ and t is in the domain of f, then $f(t) \in S$.*

If you prefer, you may use the following definition, which is equivalent to the one above.

Definition 21. *The statement that the simple graph f is **continuous** at the point $(x, f(x))$ means that if A and B are any two horizontal lines with the point $(x, f(x))$ between them, then there are two vertical lines a and b with the point $(x, f(x))$ between them such that every point on the graph f that is between a and b is also between A and B.*

Problem 43. *Show that if f is the graph which contains only the two points $(0,0)$ and $(1,1)$, then f is continuous at the point $(0,0)$.*

Problem 44. *Let f be the graph such that $f(x) = x^2$ for each number x. Show that f is continuous at the point $(2,4)$.*

Problem 45. *Let f be the simple graph such that $f(x) = 1$ for all numbers $x > 0$, and $f(0) = 0$. Show that f is not continuous at the point (0,0).*

Problem 46. *If f is a simple graph and $x_1, x_2, x_3, \ldots$ is a sequence of points in the domain of f converging to the number c in the domain of f, and f is continuous at $(c, f(c))$, then $f(x_1), f(x_2), \ldots$ converges to $f(c)$.*

Problem 47. *If f is a simple graph and $c > 0$ and g is the graph so that $g(x) = cf(x)$ for each number x in the domain of f, and f is continuous at the point $(a, f(a))$, then g is continuous at the point $(a, g(a))$.*

Problem 48. *If M is a finite subset of a segment S and p is a point of S, there is a segment T that is a subset of S and contains p but contains no point of M that is not p.*

Problem 49. *Show that if f is a simple graph and $(x, f(x))$ is a point on f, and x is not a limit point of the domain of f, then f is continuous at $(x, f(x))$.*

Question 1. *Suppose f is a simple graph with domain the interval $[-1,1]$ and $f(0) = 0$. Suppose also that f is continuous at the point $(0,0)$. Must there be two vertical lines with $(0,0)$ between them such that f is continuous at each point on f between the two vertical lines?*

Problem 50. *If f and g are simple graphs having domain the interval I, $a \in I$, $f(a) = g(a)$, and each of f and g is continuous at the point $(a, f(a))$, and h is a simple graph with domain I, and for each number x in I, $f(x) \leq h(x) \leq g(x)$ then h is continuous at $(a, h(a))$.*

Problem 51. *Let f be a graph whose domain is an interval $[a,b]$, and which has points both above and below the $x-$axis. Let c be a number $c \in [a,b]$ such that $f(c) > 0$ and let H denote the set of all numbers x in $[a,b]$ such that $f(x) \leq 0$. Show that if f is continuous at $(c, f(c))$, then c is not a limit point of H.*

Problem 52. *Let f be a graph whose domain is an interval $[a,b]$, let c be a number in $[a,b]$ such that $f(c) > 0$ and let K denote the set of all numbers x in $[a,b]$ such that $f(x) > 0$. Show that if f is continuous at $(c, f(c))$, then c is a limit point of K.*

Question 2. *If $M_1, M_2, M_3, \ldots$ is a sequence of point sets such that for each positive integer n, $M_{n+1} \subseteq M_n$, must there be a point that is common to all the sets of the sequence $M_1, M_2, M_3, \ldots$?*

Problem 53. *Let p be a point and for each positive integer n, let $S_n = (p - 1/n, p + 1/n)$. Show that every segment containing p contains one of the segments in the sequence $S_1, S_2, S_3, \ldots$.*

Problem 54. *Let p be a point and for each positive integer n, let $S_n = (p - 1/n, p + 1/n)$. Show that p is the only point common to all the segments in the sequence $S_1, S_2, S_3, \ldots$.*

Problem 55. *Assume that p is a limit point of the point set M. Show that there is a sequence of points of M all different and each different from p which converges to p.*

Problem 56. *Show that if n is a positive integer, then $2^n > n$.*

Problem 57. *Show that the sequence* $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8} \ldots$ *converges to* 0.

Question 3. *If* $S_1, S_2, S_3, \ldots$ *is a sequence of segments such that for each positive integer* n, $S_{n+1} \subseteq S_n$, *then must there be a point common to all the segments of the sequence?*

Problem 58. *Assume that* $I_1, I_2, I_3, \ldots$ *is a sequence of intervals such that for each positive integer* n, $I_{n+1} \subseteq I_n$. *Show that there is a point that is common to all the intervals of the sequence* $I_1, I_2, I_3, \ldots$.

Problem 59. *Assume that* $I_1, I_2, I_3, \ldots$ *is a sequence of intervals such that for each positive integer* n, $I_{n+1} \subseteq I_n$ *and the length of* I_n *is less than* $1/n$. *Show that there are not two points common to all the intervals of the sequence* $I_1, I_2, I_3, \ldots$.

Definition 22. *The statement that* f *is a* ***continuous graph*** *means that* f *is a simple graph which is continuous at each of its points.*

Definition 23. *The statement that the point set* M *is a* ***closed*** *point set means that if* p *is a limit point of* M, *then* p *is in* M. *An equivalent definition is that* M *is closed if and only if there is no limit point of* M *that is not in* M.

Note that this means that a set with no limit point, a finite set for example, is closed.

Problem 60. *Assume that* f *is a continuous graph with domain the interval* $[a,b]$, *and there is a point* x *in* $[a,b]$ *such that* $f(x) \geq 0$. *Show that if* M *is the set of all points* t *such that* $f(t) \geq 0$, *then* M *is a closed point. Similarly if there is a point of* f *on or below the x-axis, then the set of all such points is a closed point set.*

Problem 61. *Show that if* M *is a closed and bounded point set, then* M *has a rightmost point.*

Definition 24. *If* $p_1, p_2, p_3, \ldots$ *is a sequence then the statement that* $q_1, q_2, q_3, \ldots$ *is a subsequence of* $p_1, p_2, p_3, \ldots$ *means that there is an increasing sequence of positive integers* $n_1, n_2, n_3, \ldots$ *such that for each positive integer* i, $q_i = p_{n_i}$.

Problem 62. *Show that if* $p_1, p_2, p_3, \ldots$ *is a sequence that converges to the point* c *and* $q_1, q_2, q_3, \ldots$ *is a subsequence of* $p_1, p_2, p_3, \ldots$, *then* $q_1, q_2, q_3, \ldots$ *converges to* c.

Question 4. *Assume that* H *and* K *are closed point sets such that* $H \cup K = [0,1]$. *Must* H *and* K *have a point in common?*

Problem 63. *Assume that* f *is a continuous graph with domain the interval* $[a,b]$ *and* $f(a) > 0$ *and* $f(b) < 0$. *Show that there is a number* $c \in (a,b)$ *such that* $f(c) = 0$.

Problem 64. *Assume that f is a continuous graph whose domain includes an interval $[a,b]$, and L is a horizontal line, and $(a, f(a))$ is below L and $(b, f(b))$ is above L. Show that there is a number x between a and b such that $(x, f(x))$ is on L.*

Problem 65. *Assume that f is a graph, and $x_1, x_2, x_3, \ldots$ is a sequence of points in the domain of f converging to to the point c in the domain of f and $f(x_1), f(x_2), f(x_3), \ldots$ converges to the point y and $f(c) \neq y$. Show that f is not continuous at the point $(c, f(c))$.*

Problem 66. *Assume that f is a graph, and $x_1, x_2, x_3, \ldots$ is a sequence of points in the domain of f converging to to the point c in the domain of f, and for each positive integer n, $f(x_n) = n$. Show that f is not continuous at c.*

Problem 67. *Assume f is a graph, and $x_1, x_2, x_3, \ldots$ is a sequence of points in the domain of f converging to to the point c in the domain of f, and for each positive integer n $f(x_n) \geq n$. Show that f is not continuous at c.*

Problem 68. *Assume f is a graph, and $x_1, x_2, x_3, \ldots$ is a sequence of points in the domain of f converging to to the point c in the domain of f, and the range of the sequence $f(x_1), f(x_2), f(x_3), \ldots$ is not bounded. Show that f is not continuous at c.*

Problem 69. *Assume that f is a simple graph whose domain is the interval $[a,b]$ and f is continuous at each number in $[a,b]$. Show that the range of f is bounded.*

Problem 70. *Show that if the set M is not closed then there is a sequence of points of M that converges to a point p that is not in M.*

Problem 71. *Assume f is a continuous graph and $x_1, x_2, x_3, \ldots$ is a sequence of points in the domain of f whose range has p as a limit point. Assume that p is in the domain of f. Assume that the sequence $f(x_1), f(x_2) f(x_3), \ldots$ converges to the number c. Show that $f(p) = c$.*

Problem 72. *Assume f is a simple graph with domain $[a,b]$, and f is continuous at each number in $[a,b]$, then the range of f is a closed point set.*

Definition 25. *If f is a simple graph, the statement that the number m is the derivative of f at the number c means that*
(1) c is in the domain of f
(2) c is a limit point of the domain of f, and
(3) if (a,b) is a segment containing m, then there is a segment (h,k) containing c such that if t is a number in (h,k) and in the domain of f, and $t \neq c$, then $\frac{f(t)-f(c)}{t-c} \in (a,b)$.

Notation 8. *If f is a simple graph which has a derivative at the number c, then $f'(c)$ denotes the derivative of f at c.*

Problem 73. *If f is a simple graph with domain all real numbers and f satisfies $-x^2 \leq f(x) \leq x^2$ for each number x, then $f'(0) = 0$.*

Problem 74. *Assume f is a simple graph, and x is in the domain of f, and the derivative of f at x is m and k is a number different from m. Show that k is not the derivative of f at x. That is: show that a function f can't have two derivatives at a number x.*

Problem 75. *Assume f is a simple graph whose domain contains $[a,b]$, $x \in (a,b)$, and if $t \in [a,b]$, then $f(x) \geq f(t)$, and f has a derivative at x. Show that $f'(x) = 0$.*

Recall that if x is a number, $|x| = x$ if $x \geq 0$ and $|x| = -x$ if $x < 0$.

Problem 76. *Suppose that ε is a positive number. Show that the segment $(x-\varepsilon, x+\varepsilon)$ is the set of all points t such that $|t-x| < \varepsilon$.*

Definition 26. *The statement that the function f is continuous at the number c means that c is in the domain of f and f is continuous at the point $(c, f(c))$.*

Problem 77. *Show that the function f is continuous at the number c if and only if*
(1) c is in the domain of f and
(2) if ε is a positive number, there is a positive number δ such that if x is a number in the domain of f and $|x-c| < \delta$, then $|f(x) - f(c)| < \varepsilon$.

Problem 78. *Assume f is a simple graph, and x is in the domain of f and x has a derivative at $(x, f(x))$. Show that f is continuous at $(x, f(x))$. Hint: If you don't see how to get started on this one, then assume that $f'(x) = 0$ and show that f is continuous in this special case.*

Definition 27. *By a tangent line to a graph f at a point $(x, f(x))$ on the graph is meant a line containing $(x, f(x))$ whose slope is the derivative of f at x.*

Definition 28. *The number c is said to be a **maximum** of the simple graph f if $f(x) \leq c$ for each number x in the domain of f and $c = f(a)$ for some point a in the domain of f. The **minimum** is defined similarly.*

Problem 79. *If f is a simple graph with domain $[a,b]$ and f is continuous at every point in $[a,b]$ then the range of f is a point or an interval, and thus f has a maximum and a minimum value.*

Problem 80. *Assume f is a function with domain $[0,1]$, $f(0) = 0$, and that $f'(x) > 1$ for each $x \in [0,1]$. Show that for each x such that $0 < x \leq 1$, $f(x) > x$.*

Definition 29. *If $[a,b]$ is an interval, by a **partition** of $[a,b]$ is meant a finite set, $\{x_0, x_1, \ldots, x_n\}$ such that $x_0 = a$, $x_0 < x_1 < \cdots < x_n$ and $x_n = b$.*

We will assume that our functions have, like continuous functions, a maximum and a minimum value on each interval in their domain.

To compute a Riemann sum for a function f over an interval $[a,b]$ that is in the domain of f we first choose a partition $\{x_0, x_1, \ldots, x_n\}$ of $[a,b]$. Then for $0 \leq i \leq n-1$ we multiply the length $x_{i+1} - x_i$ of each of the intervals by the value of f at some point $t_i \in [x_i, x_{i+1}]$. Then add all these number to obtain the Riemann sum. If the graph is above the $x-$axis, each product will be the area of a rectangle.

Formally our definition is:

The statement that the number s is a **Riemann sum** for f on $[a,b]$ means that there is a partition $\{x_0, x_1, \ldots, x_n\}$ of $[a,b]$ and a set $\{t_1, t_2, \ldots, t_n\}$ of numbers such that for each integer i, $1 \leq i \leq n$, $t_i \in [x_{i-1}, x_i]$, and

$$s = \sum_{i=1}^{n} f(t_i)(x_i - x_{i-1})$$

We will be mostly interested in **upper** and **lower** Riemann sums.

Let $P = \{x_0, x_1, x_2, \ldots, x_n\}$ be a partition of [a,b] and let f be a function with domain $[a,b]$.

The statement that s is the **upper Riemann sum** for f on the partition P means that s is the Riemann sum:

$$s = \sum_{i=1}^{n} f(t_i)(x_i - x_{i-1})$$

where for each positive integer $i \leq n$, $f(t_i)$ is the maximum value of f on the interval $[x_{i-1}, x_i]$. Note that s is the largest Riemann sum over the partition P.

Similarly the **lower Riemann sum** for f on the partition P is a number

$$s = \sum_{i=1}^{n} f(t_i)(x_i - x_{i-1})$$

where for each positive integer $i \leq n$, $f(t_i)$ is the minimum value of f on the interval $[x_{i-1}, x_i]$.

$U_P(f)$ will denote the upper Riemann sum for f on the partition P and $L_P(f)$ will denote the lower Riemann sum for f on $[a,b]$ over the partition P.

Problem 81. *Let $f(x) = 0$ for each number in $(0,1]$ and let $f(0) = 1$.*
(1) Show that if P is a partition of $[0,1]$, then $0 < U_P f$, and
(2) 0 is a limit point of the set of all upper Riemann sums for f.

Problem 82. *Let f be as in the Problem 81. What is the set of all lower Riemann sums for f? What is the set of all upper Riemann sums for f?*

Show that the greatest lower bound of the set of all upper Riemann sums for f on $[a,b]$ is equal to the least upper bound of the set of all lower Riemann sums for f on $[a,b]$.

Definition 30. *If f is a bounded simple graph with domain the interval $[a,b]$ then the **upper integral** from a to b of f is the greatest lower bound of the set of all upper Riemann sums for f on $[a,b]$ and is denoted by ${}_U\int_a^b f$. Similarly, the **lower integral** from a to b of f is the least upper bound of the set of all lower Riemann sums for f on $[a,b]$ and is denoted by ${}_L\int_a^b f$.*

Definition 31. *If f is a bounded simple graph with domain $[a,b]$ then the statement that f is **Riemann integrable** on $[a,b]$ means that ${}_L\int_a^b f = {}_U\int_a^b f$ and when a function is Riemann integrable we denote the Riemann integral by $\int_a^b f$.*

Definition 32. *We define $\int_a^a f$ to be zero.*

Problem 83. *Show that if f is is a function whose domain is the interval $[a,b]$, and $f(m)$ and $f(M)$ are the smallest and largest values of $f(x)$ for $x \in [a,b]$, and $P = \{t_0, t_1, \ldots, t_n\}$ is any partition of $[a,b]$, then $U_P f \leq f(M)(b-a)$ and $L_P f \geq f(m)(b-a)$.*

Problem 84. *If f is a continuous function with domain the interval $[a,b]$, and P is a partition of $[a,b]$, then $L_P(f) \leq U_P(f)$.*

Problem 85. *If f is a continuous function with domain $[a,b]$, and for each number x in $[a,b]$, $f(x) \geq 0$, and for some number z in $[a,b]$, $f(z) > 0$ and f is continuous at z, then ${}_U\int_a^b f > 0$.*

Definition 33. *The statement that the partition Q of the interval $[a,b]$ is a **refinement** of the partition P of $[a,b]$ means that $P \subseteq Q$.*

Problem 86. *If f is a continuous function with domain the interval $[a,b]$ and P is a partition of $[a,b]$, and Q is a partition of $[a,b]$ and Q is a refinement of P, then $L_P(f) \leq L_Q(f)$, and $U_P(f) \geq U_Q(f)$.*

Problem 87. *If f is a continuous function with domain the interval $[a,b]$ then ${}_L\int_a^b f \leq {}_U\int_a^b f$.*

Problem 88. *If f is a continuous function with domain the interval $[a,b]$ and for each positive number ε, there is a partition P of $[a,b]$ such that $U_P(f) - L_P(f) < \varepsilon$, then f is Riemann integrable on $[a,b]$.*

Problem 89. *If f is a continuous function with domain the interval $[a,b]$, and ε is a positive number, then there is a partition $\{x_0, x_1, x_2, \ldots, x_n\}$ of the interval $[a,b]$ such that for each positive integer i not larger than n, if u and v are two numbers in the interval $[x_{i-1}, x_i]$, then $|f(u) - f(v)| \leq \varepsilon$.*

Problem 90. *If f is a continuous function with domain the interval $[a,b]$ then f is Riemann integrable on $[a,b]$.*

Problem 91. *If* $[a,b]$ *is an interval and* $c \in (a,b)$ *and* f *is integrable on* $[a,c]$ *and* $[c,b]$ *and* $[a,b]$*, then* $\int_a^c f + \int_c^b f = \int_a^b f$*. Hint: Show first that* $\int_a^c f + \int_c^b f \geq \int_a^b f$*.*

Problem 92. *If* f *is a continuous function with domain the interval* $[a,b]$*, then there is a number* c *in* $[a,b]$ *such that* $\int_a^b f = f(c)(b-a)$*.*

Problem 93. *If* f *is a continuous function with domain the interval* $[a,b]$ *and* F *is the function such that for each number* x *in* $[a,b]$*,* $F(x) = \int_a^x f$*, then for each number* c *in* $[a,b]$ F *has a derivative at* c *and* $F'(c) = f(c)$*.*

Problem 94. *If* f *is a function with domain the interval* $[a,b]$*, and for each number* $x \in [a,b]$ f *has a derivative* $f'(x)$*, and* f' *is continuous at each point in* $[a,b]$*, then* $\int_a^b f' = f(b) - f(a)$*.*